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Session Objective:

At the end of the session, the participants will be able to **identify** safety and health hazards in their workplace and their corresponding basic rules and control measures

The footer contains a cartoon character holding a sign that says 'Occupational Safety and Health', the text 'Department of Labor and Employment' and 'OCCUPATIONAL SAFETY AND HEALTH CENTER', and the OSHC logo. The footer is decorated with a green and blue horizontal bar.

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Outline

- Workplace Good Housekeeping (5S System)
- Fire Safety
- Machine Safety
- Safety on Material Handling & Storage
- Electrical Safety



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Safety / Direct Hazards



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Safety Hazards

- Hazards from poor housekeeping
- Fire hazards
- Machine-related / mechanical hazards; Hazardous motions
- Hazards associated with materials handling
- Electrical hazards



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Workplace Good Housekeeping



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SIGNS OF POOR HOUSEKEEPING

- Cluttered and poorly arranged areas
- Materials gathering rust and dirt from disuse
- Blocked aisleway
- Untidy or dangerous storage of materials
- Overflow storage areas and shelves
- Presence of items no longer needed or in excess
- Dusty, dirty floors and work surfaces
- Tools and equipment left in work areas
- No waste bins and containers
- Presence of spills and leaks



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ACCIDENTS FROM POOR HOUSEKEEPING

- Being hit by falling objects
- Tripping over loose objects on floors, stairs and platforms
- Slipping on greasy, wet or dirty surfaces
- Striking against projecting, poorly stacked items or misplaced material
- Cutting, puncturing, or tearing the skin of hands or other parts of the body on projecting nails, wire or steel strapping
- Electrocution from exposed live wires



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Planning a Good Housekeeping Program (5S)

- **5S** is a tool that represents the basic principles of housekeeping and workplace organization. It is more than cleaning and painting. It is a disciplined approach to keep the workplace efficient and effective.
- **Benefits Of 5S:**
 - Maintains safe and healthy work conditions
 - High morale. Employees feel good in their second home.
 - Improve company image.

5S of Good Housekeeping

5S	DESCRIPTION / ACTION	ILLUSTRATION
<u>Seiri</u> Sort Suriin	▪ identify and eliminate all unnecessary items from the workplace	
<u>Seiton</u> Systematize Sinupin	▪ Put all essential materials in a systematic order	
<u>Seiso</u> Sweep Simutin	▪ Clean and/or polish the workplace to attain a dirt or dust-free state	
<u>Seiketsu</u> Standardize Siguruhin	▪ Develop procedures, schedules, practices ▪ Regularly audit using checklists and measures of housekeeping	
<u>Shitsuke</u> Self – Discipline Sariling - Kusa	▪ A condition where all members practice the above 4S spontaneously and willingly as a way of life	

For a 5S & Housekeeping Program To Be Effective

- Management commitment and employee support
- Housekeeping & 5s policy program and procedures
- Training
- Program evaluation
- Recognition



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Fire Safety



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Fire - Rapid oxidation with the evolution of light and heat.

Fire Triangle

Fuel – can be any combustible material



Oxygen – in sufficient quantities, combine with the fuel for combustion to occur.

Heat – energy necessary to raise the temperature of the fuel and the oxygen to a point at which they will react together.



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Common sources of Heat

- Overloaded electrical system
- Open flames
- Smoking / cigarette butts
- Hot surfaces
- Friction
- Cutting / welding
- Electrical spark



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Some Facts About Oxygen

At 16% concentration, combustion is slowed down and will eventually diminish.

At 23% concentration, the atmosphere is considered oxygen enriched and fire will intensify.

Gaseous element in air at 21% by volume. It is essential for respiration..



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Fires Can Be Classified According To The Fuel It Consumes

Class	Examples	Pictogram
A – Ordinary Combustibles	paper, wood, or plastic	
B – Surface Fires	Oil, gasoline, alcohol, grease, oil – based paints	
C – Energized Electrical Equipment	Appliances, power tools, motors, electrical panels	
D – Combustible Metal	Magnesium, Sodium, Potassium	
K – Kitchen Fires	Combustible vegetable or cooking fats	



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The Burning Process (Stages of Fire Development)

PHASE	FIGURE	CHARACTERISTICS
Ignition / Incipient		Materials reach their ignition temperatures and a fire is started.
Growth		Fire begins to grow spreading chain of reaction resulting in an increase in size.
Free - Burning		All the contents within the perimeter of the fire's boundaries are burning.
Decay		Fire will extinguish itself, when the fuel or oxygen supply is exhausted.



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Importance of Fire Safety

Life Safety

The primary goal of fire safety efforts is to protect building occupants from injury and to prevent loss of life.

Property Protection

The secondary goal of fire safety is to prevent property damage.

Protection of Operations

By preventing fires and limiting damage we can assure that work operations will continue.



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Principles Of Fire Prevention And Control

Prevent the Outbreak of Fire	- Provision of Hot Work Permit System - Practice safe storage of chemicals - Never overload a power outlet - Smoke only in designated smoking areas - Correct all defective wiring systems	
Provide for Early Detection	A complete protective signaling and control system including fire detection, alarm and communication Fire Alarms, Detectors, annunciators	
Prevent the Spread of Fire	- Compartmentation - Fixed Fire Protection System - Sprinklers, Hydrants, Fire Hose/ Fire Hose Cabinets	
Provide for Prompt Extinguishment	- Removal of Fuel - Excluding or Limiting Oxygen - Cooling - By Interrupting the Chain Reaction	
Provide for Prompt and Orderly Evacuation	- Fire emergency evacuation route maps/evacuation plans - Working/active fire brigade team/ emergency response team	



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Using Portable Fire Extinguisher



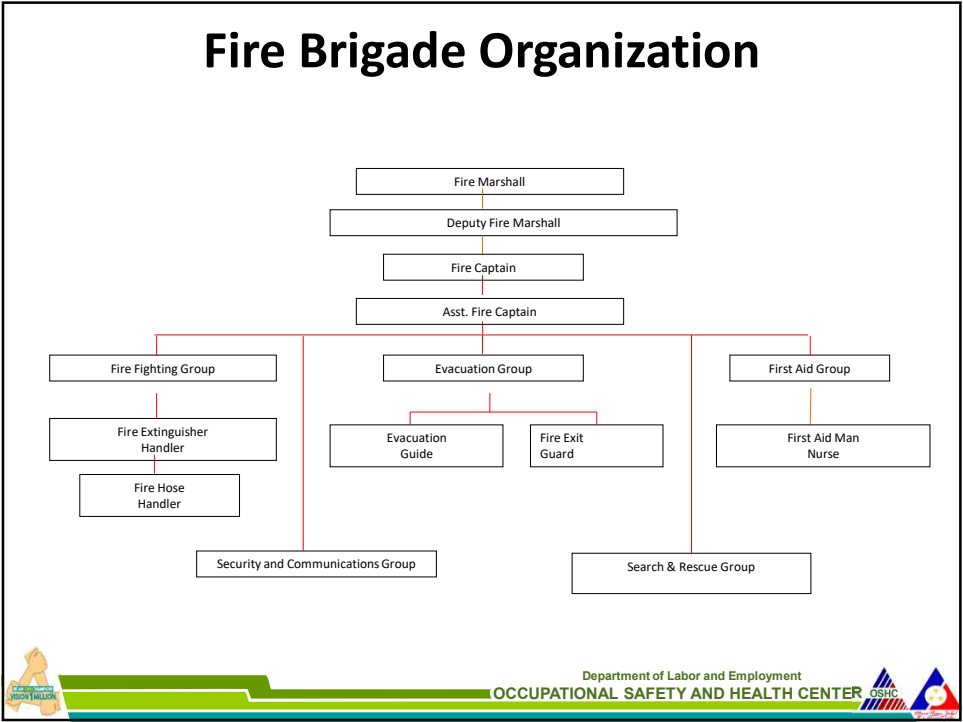
- Start approximately 6-8 feet from the fire then squeeze the trigger slowly while moving towards the fire
- 10-lb fire extinguishers lasts only 10-20 seconds. It should be used correctly.
- When the fire is extinguish, stop pulling the trigger. In the event of a re-ignition, at least there would still be contents inside the unit.
- Ensure that you have an escape path in case the fire is not extinguished



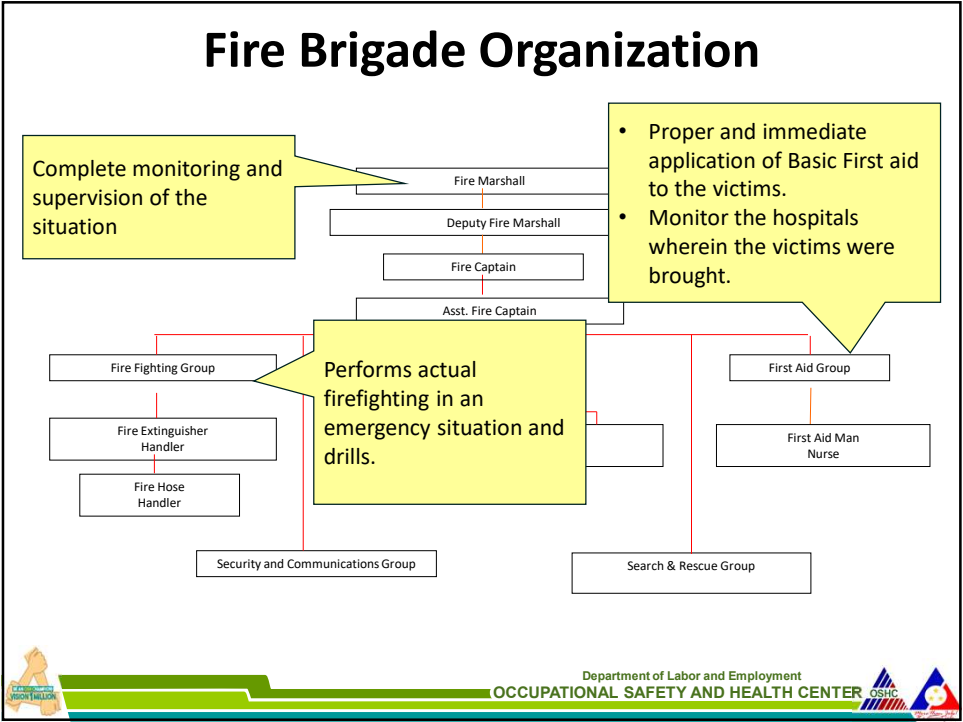
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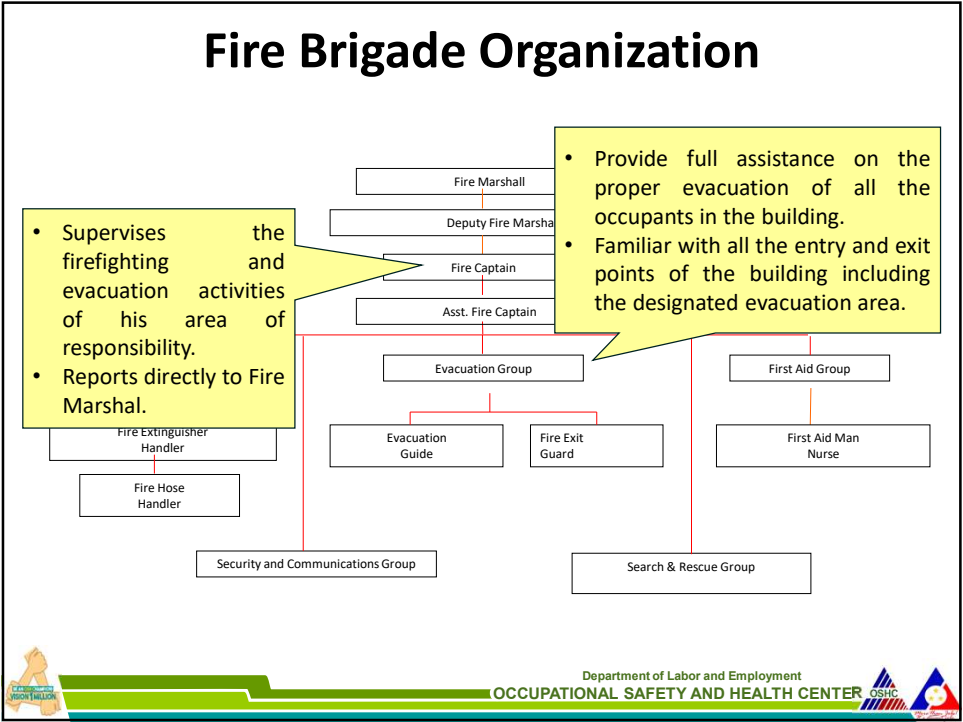
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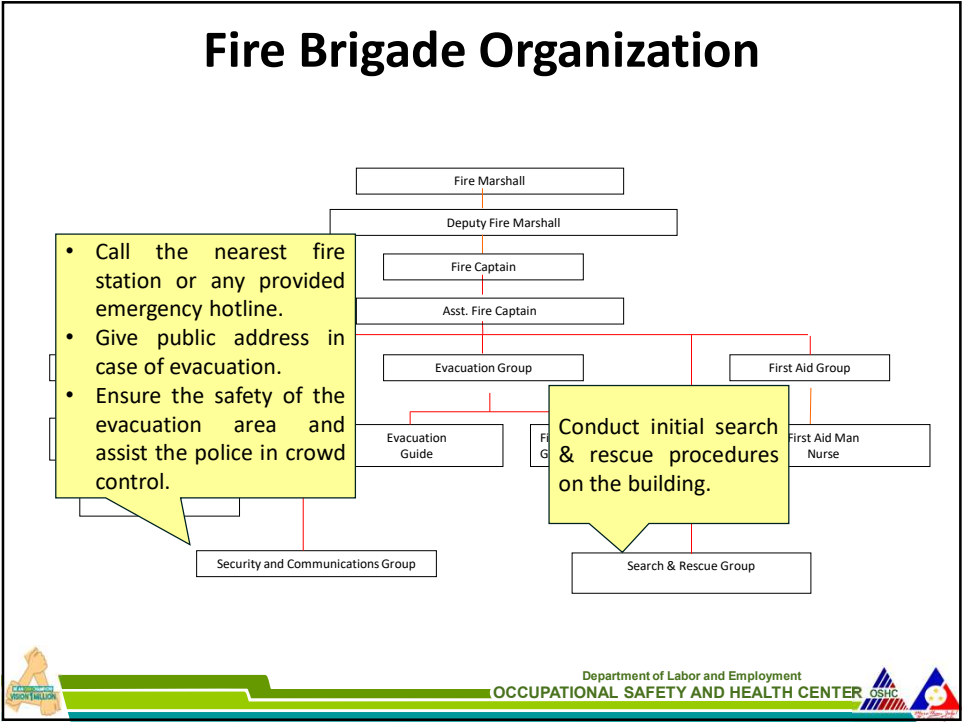
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Fire Safety Program

- Fire Detection & Alarm System
- Fire Safety Training
- First Aid Fire Protection System
- Conduct of drills
- Fixed Fire Protection System
- Working / active fire brigade team / emergency response team
- Fire Exit Doors, Fire Exit Signs, directional arrows
- Working / updated Emergency Response Plans
- Fire emergency evacuation route maps /evacuation plans
- Compliance to standards (OSHS, Fire Code)
- Inspection & maintenance system of fire detection, alarm, control & suppression systems.
- Hot work permit system



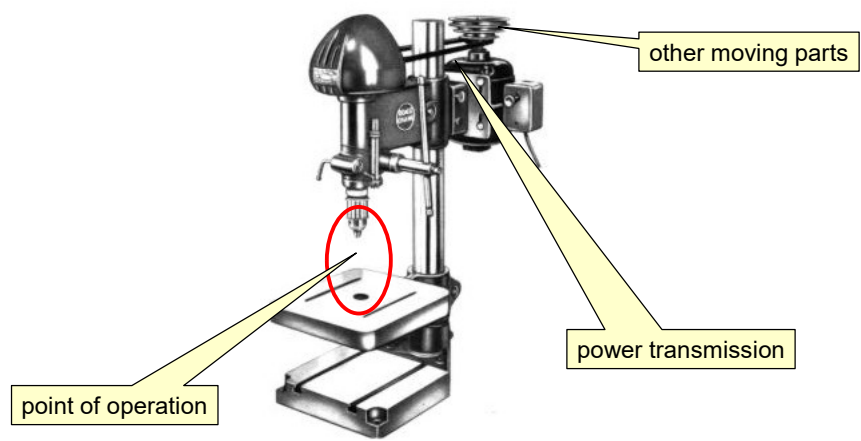
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Machine Safety

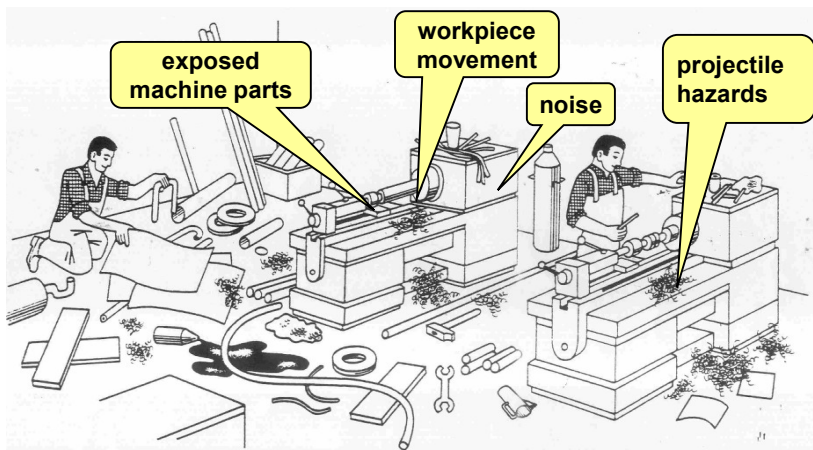


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Where Mechanical Hazards Occur



Hazards Associated With Machines



Effects Of Machine Related Accidents

- Can result to severe injuries.
- Loss of trained and skilled employee.
- Loss in productivity.
- Damaged equipment.



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Machine Safety

- Safeguarding any machine part that may cause injury. It is the prevention of accidents when working with machines.
- Prevents:
 - Loss of life
 - Severe accidents or serious injury
 - Loss of production
 - Equipment damage and repairs
 - Having time spent on accident investigation and other statutory requirements.



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Machine Guards

- Guards are barriers that prevent entry of an individual's hands or other body parts into a hazard area.
- Installed to minimize the risk of injury to machine operators or other persons from hazardous machine parts, materials being processed, or scrap.



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Requirements For Machine Guards

- Prevent employee contact with hazardous moving parts
- Secured and durable
- Prevent falling materials into moving parts
- Create no new hazards
- Must not interfere with worker productivity
- Should allow for proper and safe maintenance and lubrication



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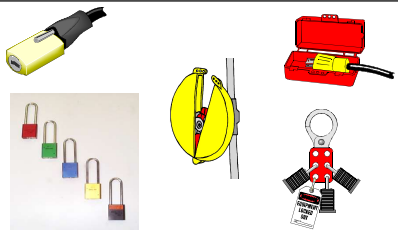
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Category Of Machine Guards

- Preventing Access
 - Fixed enclosing guards
 - Movable Guards with interlocking switches
 - Adjustable Guards
 - Manually adjustable
 - Self-adjusting
- Preventing Dangerous Motion
 - Photoelectric light curtains/Presence sensing device
 - Two hand controls
 - Pullback devices
 - Restraint device

Lock-out/Tag-out System (Loto)

The standard for the control of hazardous energy sources which covers maintenance of machines in which the unexpected start up of machines or release of stored energy could cause injury to employees.

LOCK OUT DEVICE	TAG OUT DEVICE
	

Machine Safety Program

An effective Machine Safety Program must have the following:

- Equipment/Machine inspection and maintenance program.
- Installation of safety devices (guards, relief valves, provision for LOTO, interlocks, limit switches, etc)
- Compliance to regulatory requirements and standards (e.g. mechanical permits, OSHS, etc)
- Implementation of LOTO system.
- Having a machine operation and safety procedures.
- Machine alarm and warning signal system.
- Equipment/machine safety signages.
- Application of Job hazard Analysis



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Safety In Material Handling & Storage



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Materials Handling

A technique which includes the art of lifting, placing, storing or moving of materials through the use of appropriate handling equipment and men.

Manual Handling

- is the of lifting, transporting and packaging of products using own physical strength.
- Hand operated handling, transporting and packaging of products.

Mechanical Handling

- pertains to more rigid, powered and non-powered mechanics mainly for handling bulky and heavy items



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Hazards Associated with Materials Handling

- Failure of the lifting equipment
- Falling load
- Collision
- Trapped between objects



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Manual Handling – Mistakes That Cause Injuries

- Bending Back
- Twisting with load
- Attempting too much weight (load too heavy)
- Reaching too far
- Lifting to one side
- Off – balance shifting
- Failure to use personal PPE, such as gloves, safety shoes



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Proper Procedure of Manual Lifting

REMINDERS	ACTIONS	DO WEAR...
▪ Inspect materials for sharp edges, burrs, rough or slippery surfaces. ▪ Get a firm grip on the object. ▪ Keep fingers away from pinch points, especially when setting down materials. ▪ Wipe off greasy, wet, slippery, or dirty objects before trying to handle them. ▪ Never attempt to lift that are either too heavy or bulky to handle safely. 	▪ Stand close to the load and face the way you intend to move ▪ Keep feet apart ▪ Be sure you have a good grip on the load ▪ Look forward to keep back straight ▪ Keep arms straight ▪ Tighten abdominal muscles ▪ Tuck chin into the chest	▪ lightweight, flexible, tear and puncture-resistant clothing, ▪ safety boots with toe caps and slip-resistant soles, and ▪ protective gloves, appropriate for the materials being handled. 



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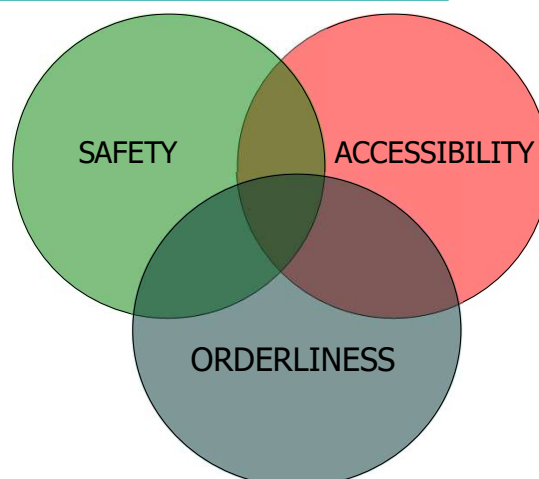
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Mechanical Handling General Requirements

- Operators must be under skills training and must be authorized.
- Equipment must be regularly inspected and maintained.



Principles Of Material Storage



Materials Storage Room General Requirements

- Stored materials must not create a hazard.
- Should be properly illuminated and ventilated.
- Materials are properly identified and labeled (including hazard labels).
- Should have proper danger or warning signs.
- Must have a smooth flow of materials, material handling equipment and people.
- Storage areas must be kept free from accumulated materials that may cause tripping, fires, or explosions, or that may contribute to the harboring of rats and other pests).
- When stacking and piling materials, it is important to be aware of such factors as the materials' height and weight, how accessible the stored materials are to the user, and the condition of the containers where the materials are being stored.
- Materials & material handling equipment should not obstruct emergency equipment such as fire alarm buttons, evacuation map, first aid kits, fire extinguishers (portable or fixed) etc.



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Electrical Safety



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Clues That Electrical Hazards Exist

- Tripped circuit breakers or blown fuses
- Warm tools, wires, cords, connections, or junction boxes
- Circuit breaker that shuts off a circuit
- Worn or frayed insulation around wire or connection



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More Frequent Accidents Happen at Low Voltage Areas

- Most employees work and move around in this area
- Not all workers have the proper trainings and experience
- It is treated as less dangerous.

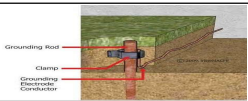
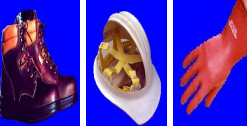


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Protection Against Hazards of Electricity

Maintain adequate grounding of circuit and equipment	intentionally creating a low-resistance path that connects to the earth	
Properly install guarding	enclosing electric equipment to make sure people don't accidentally come into contact with its live parts	
Use of adequate and approved type of personal protective equipment	- Proper foot protection (not tennis shoes) - Rubber insulating gloves, hoods, sleeves, matting, and blankets - Hard hat (insulated - nonconductive)	
Use of Lockout/Tagout in maintaining electrical equipment		
Use of Ground fault circuit interrupters (GFCI)	- Protects operator from electric shock - Detects difference in current going into the equipment and returning from it - If ground fault detected, GFCI shuts off electricity in 1/40th of a second	



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Review



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Review questions

1. Electricity-related accidents happen most frequently in:
a. Low voltage area b. High voltage area
2. Enumerate the components of the 5S of Good Housekeeping.
3. Identify the three basic elements of fire.
4. _____ are means of effectively preventing the person/worker from coming into contact with the moving parts of the machine or equipment
5. Safety, orderliness and accessibility are the main components of _____.



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Clear Points



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Clear Points

- **5S** is a tool that represents the basic principles of housekeeping and workplace organization.
- The principles of fire prevention and control are:
 - Prevent the Outbreak of Fire
 - Provide for Early Detection
 - Prevent the Spread of Fire
 - Provide for Prompt Extinguishment
 - Provide for Prompt and Orderly Evacuation
- Guards are barriers that prevent entry of an individual's hands or other body parts into a hazard area.
- Operators must be under skills training. Storage areas shall be safe, accessible and orderly as well.



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Clear Points

- Regular inspection of electrical fixtures and equipment/machines will allow early detection and correction of defects.
- All protective equipment shall be of approved design and construction appropriate for the exposure and the work to be performed.
- To perform an accurate and completed JHA, you need to:
 - Select the job to be analyzed
 - Observe the job as it is performed to break it down into basic steps
 - Analyze each step to determine existing and potential hazards associated with each step
 - Develop recommendations for eliminating or controlling the hazards associated with each step



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